Tree-like Causal Latency Model (TCLM) and Dual-Pulse Synchronization Protocol (DPSP): From Observational-Delay Deconstruction to Interplanetary and Cross-Galactic Deep-Space Chronometric Engineering

> **Language Disclaimer** In the event of any discrepancy or inconsistency between this English version and the original Chinese version, the Chinese version shall prevail.

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> **Manifesto** Newton was right, and Einstein was also right.
> The causal tree is a framework that allows both to be true simultaneously.

Abstract

This paper points out that equating the mathematical boundary failure of the Lorentz transformation at $v > c$ with "time reversal" misplaces observational signal propagation latency as a reversal of physical time. We argue that using velocity breakthrough as a sufficient condition for returning to the past is invalid in operational definition; its geometric limit is propagation latency compressed to zero (Ping=0), not negative latency ($L < 0$). This paper makes no universal claim about whether "returning to the past" is physically possible; its scope is limited to the operational definition of velocity superposition as an implementation means.

To this end, we propose the Tree-like Causal Latency Model (TCLM), which strictly decouples time into a unidirectionally settled Topological Causal Order (Trunk $\mathcal{C}_{\text{trunk}}$, $\text{sgn}(\mathcal{C}) \equiv +1$) and a Geometric Time-Rate Difference (Branches) describing local proper-time rate and observational latency.

To advance engineering, we propose a conceptual application-layer synchronization protocol, the Dual-Pulse Synchronization Protocol (DPSP), establishing Decentralized Initial-Constant Locking and Local Rate Measurement. The protocol decouples the local time-rate factor (Γ) and the initial integration constant (t_0): the local end solves branch curvature via multi-baseline vector triangulation, while the deep-space end requires only a single initial handshake to lock t_0; with Adaptive Pulse Frequency Scaling, Continuous Curvature Quadratic Compensation, and Causal Backpressure Buffering, it independently reconstructs the causal order relative to the handshake ($\mathcal{C}_{\text{trunk}} - t_0$) via worldline path integration in dynamic gravity wells.

In the Earth-Mars calibration case, after multi-frequency dispersion compensation and ephemeris constraints, the engineering alignment error-budget target is nanosecond-level ($\mathcal{O}(10^{-9}\ \text{s})$), with theoretical resolution upper-bounded by optical lattice clock hardware limits ($10^{-18} \sim 10^{-19}$). The Earth-Mars case is a theoretical calibration pending end-to-end link simulation and empirical validation; the architecture is logically extensible to interstellar scales, with engineering feasibility and error budgets remaining for future work. The model fully

respects tensor covariance and the no-communication theorem. This paper does not claim gravitational-wave detection capability, nor does it claim any FTL mechanism is engineering feasible.

Introduction and Epistemological Disclaimer

This theory employs information-theoretic physics and topological geometry for the structural description of causal chains in physical spacetime. The abstract concepts used in this paper, such as "static topological full map," "restricted read order," and "information propagation latency," are strictly confined to the domain of information theory and mathematical models, aiming to provide an objective deconstruction of physical observations without making any metaphysical assumptions or extensions regarding the ontological entity of the universe.

Chapter 1: From "Faster-than-Light Time Travel" to the Distinction Between Causal Order and Time

The core intuition of this paper can be illustrated by thunder and lightning.

The existence of thunder and the lightning we observe are two different things.

If we perceive the existence of thunder in some way before the lightning arrives, does that mean we have traveled back in time?

No. It simply means that the latency of our perception method is shorter than that of the lightning. The event itself has not reversed; only the signal latency has been shortened.

Because there is only one thunder event, it occurred. But it takes time for the lightning to travel from the location of the thunder to our location. Therefore, the lightning we see was emitted by the thunder at a past moment. What we see is always the past lightning, not the thunder itself.

If we treat the "order of seeing the lightning" as the "order of the thunder occurring," we would falsely assume the occurrences have a sequence. But if we clearly distinguish between the "existence of the thunder itself" and the "arrival time of the lightning," we realize: there is only one thunder; only the arrival times of the lightning differ.

This distinction is the first cornerstone of TCLM. This paper will formally define this distinction in Chapter 2. Let us first start with a more fundamental question to explain why this distinction is important.

Are "the order of observation" and "the order of occurrence" the same thing?

If they are not the same thing, then the claim of "faster-than-light going back in time" needs to be re-examined. The result of this examination is: going back in time faster than light seems possible only because "observation latency" and "event occurrence" are conflated. The Lorentz transformation yields imaginary numbers when $v > c$. The standard formulation in physics textbooks is "no real solution, hence unreachable for particles with mass." If this complex output is directly interpreted as "time flowing backward," it confuses two concepts.

But this misinterpretation happens to open a door: if causal order and time are not the same thing, then what are they respectively? What is their relationship?

This is the starting point of the Tree-like Causal Latency Model (TCLM). Below, we will first explain the mathematical boundary failure of the Lorentz transformation at extreme speeds, and then explain how this boundary leads to the distinction between causal order and time.

1. The Mathematical Boundary Failure of the Lorentz Transformation at Extreme Speeds

Traditional special relativity faces two core mathematical collapses when dealing with extreme speeds, both rooted in "forcing the electromagnetic propagation limit (c) to be equivalent to the ontological time of the universe":

1. **Observed Time (t') Exceeding the Real Domain**: In the transformation equation $t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$, when $v > c$, the denominator $\sqrt{1 - \frac{v^2}{c^2}}$ becomes imaginary, pushing the entire t' into the complex domain. Whether the numerator is negative depends on specific values. If this complex output is directly interpreted as "time flowing backward," it is fundamentally a misjudgment of invalid values given by the formula beyond its real domain as an actual reversal of physical time.
2. **Proper Time (τ) Exceeding the Real Domain**: In the proper-time formula $\tau = t\sqrt{1 - \frac{v^2}{c^2}}$, when $v > c$, an imaginary number i is calculated. This is the mathematical model exceeding its real domain (undefined over the real domain), not the physical structure of time turning into nothingness.

2. Proposition: Convergence and Boundary Closure of Spacetime Signal Propagation Latency Limit

2.1 Traditional Factor Domain Overflow

As above, when $v > c$, both observed time t' and proper time τ exceed the real domain; this is mathematical model failure, not physical time turning into nothingness.

2.2 TCLM Geometric Latency Rate (η) Mapping and Absolute Convergence

Define medium latency rate $\eta \equiv 1/v$, where vacuum electromagnetic latency rate is constant $\eta_0 \equiv 1/c$. Propagation latency time $L_{\text{propagation}}$ over distance Δx is:

$$L_{\text{propagation}}(\eta) = \eta \cdot \Delta x$$

When considering signal propagation beyond electromagnetic medium limit (i.e., latency compression $\eta < \eta_0$):

$$\lim_{\eta \to 0^+} L_{\text{propagation}}(\eta) = \lim_{\eta \to 0^+} (\eta \cdot \Delta x) = 0$$

- **Physical limit boundary**: Absolute lower bound of propagation time cost is exactly 0 (i.e., zero-latency state, Ping=0). This state represents complete synchronization of transmitter and receiver on trunk causal order $\mathcal{C}_{\text{trunk}}$.
- **Causal topological closure**: Domain is strictly limited to non-negative closed interval $L_{\text{propagation}} \in [0, \infty)$, $L < 0$ undefined. This is not arbitrary, but from operational requirement that propagation latency as non-negative physical quantity — latency can approach zero (Ping=0), but cannot be negative. Since limit is approached from positive side $\eta \to 0^+$, the mathematical path to $L < 0$ is completely sealed topologically and measure-theoretically.

> **On applicability boundary of relativistic factor**: Lorentz/optical factor is defined only when $\eta \geq \eta_0$ (i.e., $v \leq c$); when entering superluminal/latency-compression domain $\eta < \eta_0$ (i.e., $v > c$), model no longer uses optical factor, directly switches to real mapping $L_{\text{propagation}}(\eta) = \eta \cdot \Delta x$, and derives geometric limit $\lim_{\eta \to 0^+} L = 0$.

2.3 Conclusion and Scope Limitation

Traditional framework maps speed v onto open interval $[0, \infty)$, encountering unavoidable infinite singularity at $v = c$; TCLM via dual transformation maps physical process onto closed compact interval $\eta \in [0, \eta_0]$ (defining $L(0) = 0$ as continuous extension of $\eta \to 0^+$).

> **Absolute endpoint of latency compression is $\eta = 0$. Boundary value 0 is geometric limit. For the path of "using velocity superposition as a means", no operational-definition space crossing to negative domain exists; time-travel paradox is resolved at the domain boundary of this path. This does not constitute a universal negation of whether "time reversal is ontologically possible." Other mechanisms (e.g., closed timelike curves constructed from extreme metrics in general relativity) are outside the scope of this section.**

This paper makes no universal claim about whether "returning to the past" is possible. It only argues: using "superimposing speed beyond c" as a means alone cannot operationally achieve $L < 0$, because the required $L < 0$ is undefined in this model's domain. Whether other mechanisms (e.g., closed timelike curves constructed from extreme metrics in general relativity) are possible is outside the scope of this paper.

Falsifiability Statement

The falsification condition of this model on the velocity-superposition path is: if there exists a concrete physical mechanism by which velocity superposition can achieve $L < 0$ (signal arrival before emission), then this model's inference on that path is falsified.

This model does not claim that causal reversal is impossible. This paper claims only that, on the velocity-superposition path, the operational domain of L is $[0, \infty)$, and $L < 0$ is undefined. Whether other mechanisms can achieve causal reversal is outside the scope of this paper.

This falsification condition is not arbitrarily set. $L < 0$ corresponds to "signal arrival before emission," which is a contradictory statement in operational definition. Under the current physical framework, no known mechanism can achieve this state. If such a mechanism emerges in the future, this model is falsified.

3. From Boundary Failure to the Distinction Between Causal Order and Time

Scope Limitation: This paper's object is limited to the path of "achieving $L < 0$ via velocity superposition to $v > c$ as a time-machine mechanism" within the special-relativistic framework. Closed timelike curves (CTC, e.g., Gödel, Kerr, wormholes) constructed from extreme metrics in general relativity belong to different mechanisms, requiring separate modeling and are outside the scope of this paper. The so-called "resolution" in this paper only refers to closing the operational definition of this velocity path.

This mathematical boundary failure leads to a more fundamental distinction:

The order in which events occur (causal order) and the order in which an observer receives signals (observed time) are not the same thing.

"Faster-than-light going back in time" appears possible precisely because these two are conflated. If they are separated, FTL no longer implies going back in time, but simply a reorganization of observation latency.

This distinction is the first cornerstone of TCLM.

Scope of this Paper

This paper does not conduct a systematic literature review. Existing concepts mentioned herein are solely used to illustrate the boundaries of this paper's framework and do not constitute a complete citation or evaluation of related literature. This paper does not claim to have pioneered the concept of "quantum-entanglement timestamps." This paper treats quantum-entanglement measurement events as a **candidate source** (currently envisioned) capable of providing local timestamps. Other local independent reference events satisfying the same three operational conditions (decoupled from spatial propagation path, sufficiently stable, significantly smaller than optical latency), such as a second independent atomic-ensemble trigger event, are equivalent candidates. DPSP only uses the operable delay difference $\Delta\tau_{\mathrm{delay}} = t_{\mathrm{light}} - t_{\mathrm{ref}}$ and does not depend on whether the physical mechanism is quantum entanglement. **The establishment of this protocol does not depend on whether any specific physical mechanism is quantum entanglement; it suffices that the local reference event satisfies three operational conditions (decoupled from spatial propagation path, sufficiently stable, significantly smaller than optical latency).** This protocol does not rely on the interpretation of "whether entanglement is theoretically synchronous," but only uses the ex-post operable delay-difference measurement $\Delta\tau_{\mathrm{delay}} = t_{\mathrm{light}} - t_{\mathrm{ref}}$; whether it can effectively separate spatial latency from time-flow rate depends on the empirical stability condition (see Chapter 3, Section 1.2). This paper does not claim to have verified its practicality.

Chapter 2: Core Postulates and Global Causal Equation of the Tree-like Causal Latency Model (TCLM)

Symbol Conventions:

- Event e: a concrete event occurring on a worldline
- Trunk mark $\mathcal{C}_{\mathrm{trunk}}(e)$: projection notch of event e on the trunk, second-parameterized, only differences meaningful
- Branch proper-time mapping $\tau_{\mathrm{branch}}(\mathcal{C}_{\mathrm{trunk}}(e))$ or abbreviated $\mathcal{T}_{\mathrm{branch}}(e)$: mapping trunk mark to proper-time reading of that worldline
- Trunk growth relation denoted as $\mathcal{C}_{\mathrm{trunk}} \gg e$: denotes that event e is a link in the causal nesting. One cause triggers one or several effects, and such an effect will itself become another cause, forming a recursive causal chain.
- **Postulate 1: Global Causal Order and Branch Wave Sources**
    - **Global Causal Trunk ($\mathcal{C}_{\mathrm{trunk}}$)**: The strict total-order parameter to which all physical events ultimately belong. $\mathcal{C}_{\mathrm{trunk}}$ satisfies $\frac{d\mathcal{C}_{\mathrm{trunk}}}{d\tau} > 0$ and $\mathrm{sgn}(\Delta\mathcal{C}_{\mathrm{trunk}}) \equiv +1$. Whenever $\mathcal{C}_{\mathrm{trunk}}$ appears in the formulas of this paper, it refers to the second-parameterized version normalized by order-preserving affine transformation $\mathcal{C}_{\mathrm{trunk}} \mapsto a\mathcal{C}_{\mathrm{trunk}} + b$ ($a > 0$), with seconds as the unit. Its absolute zero point t_0 has no physical meaning; only the difference $\Delta\mathcal{C}_{\mathrm{trunk}}$ has physical meaning.

- **Global Simultaneity Definition**: If $\mathcal{C}_{\text{trunk}}(A) = \mathcal{C}_{\text{trunk}}(B)$, events A and B are said to be globally simultaneous in the trunk sense. This simultaneity is a trunk equivalence class, not presupposing any rigid simultaneity slice built into the universe, and does not imply causal reachability. Quantum-entanglement measurement pairs can satisfy $\mathcal{C}_{\text{trunk}}(A) = \mathcal{C}_{\text{trunk}}(B)$ without causal connection.

Time (proper time) is inherently a local concept; there is no "time" reading that can be directly shared across systems. Observers can only reconstruct $\mathcal{C}_{\text{trunk}}$ ex-post via propagation latency $\Delta t^{(k)}_{\text{latency}} \geq 0$ and local clock records.

Analogy: The position of the sun in the sky is not time itself; it is a physical state. A sundial maps this physical state into a time reading. Similarly, $\mathcal{C}_{\text{trunk}}$ itself is not time; it is a causal order. $\tau_{\text{branch}}(\mathcal{C}_{\text{trunk}})$ is the function mapping this causal order to the branch proper time. What we can actually reconstruct is the integral quantity relative to the initial handshake constant t_0.

- **Geometric Relative Branches and Curvature**: Represents the relative observation trajectories and local proper-time rates (τ) of physical entities growing out of the causal trunk. The essence of relative time-flow rates is the curvature of the branches on a four-dimensional geometric manifold. Relativistic effects measured by local observers (e.g., time dilation, gravitational redshift) are all results of path integrals on the geometric branches.

- **Dynamic Environmental Adaptability (Continuous Curvature-Field Fluctuation)**: Physical entities are not locked into the hierarchical structure of specific celestial bodies. Spacetime curvature is intrinsically a continuously floating field, fluctuating dynamically with the current gravitational potential and velocity field. When an object moves across galaxies, its ontological branch directly adapts to the local continuous geometric curvature. Between all objects, there only exists a "relative time difference induced by current geometric curvature," while they mutually share the exact same global causal order.

- **Forward Growth Constraint**: Under the causal-direction constraint ($\text{sgn}(\Delta\mathcal{C}_{\text{trunk}}) \equiv +1$, i.e., $\frac{d\mathcal{C}_{\text{trunk}}}{d\tau} > 0$), the growth vectors of all branches are strictly one-way downstream. The causal direction is always positive and absolutely prohibits reverse growth back upstream to the trunk within a single manifold.

Epistemological Note on "Now" and the Sense of Simultaneity

The subjective feeling of "now" and "being in the same moment as the other party" is the result of the local nervous system integrating short-latency signals, not a globally given physical simultaneity. All perceived information carries a propagation latency and is intrinsically a reconstruction of the past. At everyday scales, where light latency is extremely short and relative speeds are very low, this approximation is valid; once the distance increases or relative motion becomes significant, the approximation fails. What can be objectively shared physically and reconstructed consistently is solely the causal sequence of events.

- **Postulate 2: Decoupling of Event Ontology and Signal Carrier (The Isomorphic Model of Thunder and Lightning)**

  - **Decoupling of the Event Ontology and Lightning**: Using the "existence of the thunder/lightning event ontology" and "the lightning propagating outward" as an isomorphic

model: the instant a thunder event is settled on the topological trunk ($\mathcal{C}_{\text{trunk}}$) is the settlement point of that physical entity in the topological causal order. The lightning received by the observer is merely the geometric branch signal propagating outward from that event ($t_{\text{observed}} = \tau_{\text{branch}}(\mathcal{C}_{\text{trunk}}) + \Delta t_{\text{latency}}$).

- **Limit of Propagation vs. Time Ontology**: Lightning is only a carrier for transmitting information (geometric branch); its propagation latency does not equal the occurrence time of the thunder event itself. Even if one commands a propagation method exceeding the speed of light (perceiving the thunder event faster than lightning), it merely means the signal-transmission latency has been shortened (flattening the branch); it absolutely does not mean the observer has traveled back before the thunder event occurred (causal settlement).

2. Global Causal Equation and Wave-Source Broadcast

TCLM adopts a broadcast view of causality: an event e_A at its settlement instant broadcasts to all receivers in its causal future.

Causal future: the set of events reachable from event e_A through any causal chain. This set is determined by the one-way property of $\mathcal{C}_{\text{trunk}}$, independent of any propagation-speed assumption.

Definition 1 (Propagation Latency): Propagation latency of information carrier along path $\mathcal{P}_k$ is defined as

$$\Delta t_{\text{latency}}^{(k)} \equiv \Delta t_{\text{prop}}^{(k)} \equiv \int_{\mathcal{P}_k} \frac{\sqrt{g_{ij}^{\text{branch}} dx^i dx^j}}{v_{\text{signal}}(x)} \geq 0$$

where $v_{\text{signal}}(x) > 0$. The so-called "causal wave" in this model refers to signal carrier in operational sense, not a new physical field, its geometric limit is $L^{(k)} \equiv \Delta t_{\text{latency}}^{(k)} \in [0, \infty)$, $L < 0$ undefined, Ping=0.

Definition 2 (Arrival Equation / Wave Arrival): Let event e_A be wave source, its proper time is $\mathcal{T}_{\text{branch}}(e_A) = \tau_{\text{branch}}(\mathcal{C}_{\text{trunk}}(e_A))$, then arrival time of k-th reception event is

$$t^{(k)} = \mathcal{T}_{\text{branch}}(e_A) + \Delta t_{\text{latency}}^{(k)}, \quad \Delta t_{\text{latency}}^{(k)} \geq 0$$

After landing, new trunk mark $\mathcal{C}_{\text{trunk}}(e_{B_k})$ is reconstructed via integration, that mark corresponding to new branch becomes new wave source. Note: $\tau_{\text{branch}}^{-1}$ is not pointwise invertible; the actual reconstruction is completed via DPSP integration.

Inference: $K = N + M$ is total broadcast listeners, where N is collided targets, M is observers. All received by receivers are "effects." Receiving information itself is a kind of effect, only differing in source being direct or indirect:

- **Direct Effect**: The receiver directly interacts with e_A's signal. For example, ball B is hit by ball A, or person A directly sees ball A move.
- **Indirect Effect**: The receiver receives a signal emitted by "another effect." For example, person B hears person A shout "everyone look"; person B's information comes from person A, not ball A.

Hence no need to distinguish "observation" from "collision", uniformly denoted as $t^{(k)}$, $k = 1, \ldots, K$.

Chain (Relay): $e_A \xrightarrow{\Delta t_{A\to B}^{\text{latency}}} e_B \xrightarrow{\Delta t_{B\to C}^{\text{latency}}} e_C$, reduced in same gravity field to $t(e_C) = \mathcal{T}_{\text{branch}}(e_A) + \sum \Delta t_{\text{latency}}$, where $\mathcal{C}_{\text{trunk}}(e)$ is trunk mark of each event.

In same gravity field: $t(C) = \tau_{\text{branch}}(\mathcal{C}_A) + \Delta t^{\text{latency}}_{A\to B} + \Delta t^{\text{latency}}_{B\to C}$

Operational Definition: How to measure?

$\mathcal{C}_{\text{trunk}}$ cannot be directly measured; DPSP actually measures interval ratio of two local clocks:

$$\Gamma_{\text{local}} \equiv \frac{\Delta\tau_{\text{local}}}{\Delta\tau_{\text{base}}}$$

where $\Delta\tau_{\text{local}}$ is proper-time interval of local clock between two pulses, $\Delta\tau_{\text{base}}$ is corresponding interval of base clock. This ratio is directly measurable dimensionless quantity.

- g^{branch}_{ij} is branch spatial metric tensor, describing geometric correction of local spatial curvature to propagation path
- $\Gamma_{\text{trunk,local}} = d\mathcal{C}_{\text{trunk}}/d\tau_{\text{branch}} = \sec\theta \geq 1$ is dilation factor of trunk relative to local proper time
- $\Delta t^{(k)}_{\text{latency}} \geq 0$, $L^{(k)} \equiv \Delta t^{(k)}_{\text{latency}} \in [0,\infty)$, $L < 0$ undefined

Parameter Definitions: $t^{(k)}$ is arrival time of k-th reception event, $\mathcal{C}_{\text{trunk}}(e_{B_k})$ is reconstructed trunk mark after landing, $\mathcal{T}_{\text{branch}}(e_A)$ is wave-source proper time, $\mathcal{P}_k$ is k-th propagation path.

Causal Order $A <_{\text{causal}} B$ (Has order, satisfying reachability condition)

A can influence B iff:

1. $\mathcal{C}_{\text{trunk}}(A) < \mathcal{C}_{\text{trunk}}(B)$
2. There exists a propagation path $\mathcal{P}_k$ such that $\Delta t^{(k)}_{\text{latency}} \geq 0$ satisfies $L^{(k)} \geq 0$

Direct Influence vs Indirect Influence:

1. **Direct Influence**: Upon landing, $t^{(k)} = \tau_{\text{branch}}(\mathcal{C}_{\text{trunk}}(e_A)) + \Delta t^{(k)}_{\text{latency}}$ alters the subsequent evolution of the target system, generating a new trunk mark $\mathcal{C}_{\text{trunk}}(e_{B_k})$. If $\Delta x > c\cdot\Delta\mathcal{C}_{\text{trunk}}$ then $L < 0$ is undefined, and direct influence is impossible
2. **Indirect Influence**: $t^{(k)} = \tau_{\text{branch}}(\mathcal{C}_{\text{trunk}}(e_A)) + \Delta t^{(k)}_{\text{latency}}$ lands and only changes knowledge state of observer, its subsequent new branch is locally generated by observer, essentially series $e_A \xrightarrow{\text{obs}} O \xrightarrow{\text{causal}} e_C$

Physical causal influence: $t^{(k)}$ landing changes subsequent evolution of target system, generating new $\mathcal{C}_{\text{trunk}}(B_k)$ Informational influence: $t^{(k)}$ landing only changes knowledge state of observer, does not directly rewrite remote trunk, subsequent new branch is locally generated by observer

Conclusion: Observation and collision are operationally both arrival events. Direct influence refers to single-hop reachable path without relay, indirect influence refers to multi-hop path via observer relay. When $L < 0$ undefined, single-hop direct influence impossible, only relay-required indirect path remains.

Note: If A and C are spacelike-separated, and there exists an intermediary event B causally connected to both, then A and C, though not directly causally connected, each still have determinate positions in the global causal structure. "No direct causal relation" is not equivalent to "no causal relation."

Where $\Delta\tau_{\text{local}}$ is the interval measured by the local clock between two events, and $\Delta\tau_{\text{base}}$ is the interval measured by the base clock between the same two events. This ratio is directly measurable and can be greater

than, equal to, or less than 1. $\mathcal{C}_{\text{trunk}}$ is a post-calculated reconstructed integral quantity; its full form is in Chapter 3, Section 2.

(Note: This formula accounts for the spatial geometric-latency part; the complete time rate must incorporate gravitational redshift, detailed in Chapter 4 calibration.)

The microscopic geometric-field latency quantity is defined as:

$$\Delta t_{\text{latency}} \equiv \int_{\mathcal{P}} \frac{\sqrt{g_{ij}^{\text{branch}} dx^i dx^j}}{v_{\text{signal}}(x)}$$

(The causal-order parameter $\mathcal{C}_{\text{trunk}}$ satisfies strict monotonic increment; the integral term is comprehensively determined by geometric distance, propagation speed, and branch curvature, and is always positive, measured in seconds. In engineering implementation, this geometric latency is mapped locally as the difference between classical optical timestamps and local reference-event timestamps (currently envisioned as quantum-entanglement measurement events); see Chapter 3, Section 1.3.)

Parameter Symbol Explanations:

- t_{observed}: The scalar time when the observer receives the information residue in a specific coordinate system.
- $\mathcal{C}_{\text{trunk}}$: The underlying causal-settlement scalar of the event's occurrence, satisfying the strict monotonic-increment postulate (monotonically increasing along the proper time τ of any physical worldline, i.e., $\frac{d\mathcal{C}_{\text{trunk}}}{d\tau} > 0$). It represents the topological causal position of the event, not a globally synchronized physical clock.
- $\tau_{\text{branch}}(\mathcal{C}_{\text{trunk}})$: The proper time on the branch corresponding to the causal order $\mathcal{C}_{\text{trunk}}$.
- $\Delta t_{\text{latency}}$: The geometric latency accumulated by information propagating in the branch-space manifold, in seconds (s), strictly positive ($\Delta t_{\text{latency}} \geq 0$). This is a geometric definition; its operational definition is the local clock difference $\Delta\tau_{\text{delay}} = t_{\text{light}} - t_{\text{ref}}$, where t_{ref} is the local reference-event timestamp (currently envisioned as quantum-entanglement measurement event, mechanism different from optical latency), see Chapter 3, Section 1.3.
- $\mathcal{P}$: The physical geometric path traversed by the information carrier in the 3D branch-space manifold.
- g_{ij}^{branch}: The positive-definite geometric tensor describing local spatial curvature, gravitational potential, and information-propagation limit resistance ($i, j \in \{1, 2, 3\}$).
- dx^i, dx^j: The infinitesimal coordinate displacement of the information carrier advancing along the spatial propagation path.
- $v_{\text{signal}}(x)$: The effective propagation rate of the information carrier at coordinate point x (e.g., effective light speed of EM waves in a local field, satisfying $v_{\text{signal}}(x) > 0$), ensuring the formula possesses rigorous time dimensions $[\text{m}]/[\text{m/s}] = [\text{s}]$.

All references to "causal-order reconstruction" in this paper refer to this second-parameterized quantity.

Intuitive Understanding: The Inaccessibility of $\mathcal{C}_{\text{trunk}}$

Both $\mathcal{C}_{\text{trunk}}$ and $\tau_{\text{branch}}(\mathcal{C}_{\text{trunk}})$ are directly inaccessible. $\mathcal{C}_{\text{trunk}}$ can be likened to the order represented by a still water surface, while τ_{branch} is the proper time mapped from that water surface onto the observer's branch. However, the observer can never directly touch the still water surface:

- The water itself is a continuous fluctuating field (dynamic $\Gamma_{\text{local}}(t)$); it is always flowing.

- The act of measuring, like touching water with a ruler, inevitably causes ripples (introducing $\Delta t_{\text{latency}}$).

Therefore, what the observer can touch is always "flowing water surface + ripples" (t_{observed}), not the still water surface itself. $\mathcal{C}_{\text{trunk}}$ is the order represented by this limit; it exists independently of the water and ripples but cannot be directly touched.

Although inaccessible, it can be indirectly reconstructed via the "interval ratio":

From the operational definition above, Γ_{local} is the interval ratio directly measured locally by DPSP.

$\mathcal{C}_{\text{trunk}}$ is the post-calculated reconstructed integral:

$$\mathcal{C}_{\text{trunk_reconstructed}}(\tau) = t_0 + \int_{\tau_0}^{\tau} \frac{1}{\Gamma_{\text{local}}(\tau')}\, d\tau'$$

The sequence is: measure the interval ratio first, then integrate to reconstruct the causal order. It is not defining a causal order first and then calculating the interval ratio. This assumes the initial-handshake reference is normalized to $\Gamma_{\text{trunk,base}} = 1$; if the reference is not curvature-free, the full form must include the $\Gamma_{\text{trunk,base}}$ factor, written as $\int \frac{\Gamma_{\text{trunk,base}}}{\Gamma_{\text{local}}(\tau')} d\tau'$, see Chapter 3, Section 2 and 1.4.

If using the geometric-projection definition $\Gamma_{\text{trunk}} = \sec\theta \geq 1$, then $d(\mathcal{C}_{\text{trunk}} - t_0)/d\tau = \Gamma_{\text{trunk}}$.

Premise: This reconstruction involves the inherent latency of a second set of local events (currently envisioned as quantum-entanglement events). This latency is not a necessary condition for the reconstruction to be valid; if it is zero, $\mathcal{C}_{\text{trunk_reconstructed}}$ directly corresponds to $\mathcal{C}_{\text{trunk}}$; if it is a tiny non-zero value, there exists an unknown constant offset between $\mathcal{C}_{\text{trunk_reconstructed}}$ and $\mathcal{C}_{\text{trunk}}$, which only affects the absolute value but not the chronological order. Under current optical-lattice-clock precision, this latency is indistinguishable. The core condition for DPSP is not whether this latency is zero, but whether it is affected by different factors than optical latency, whether it is stable enough, and whether it is significantly smaller than optical latency; see Chapter 3, Section 1.2.

Analogy: The Sundial and τ_{branch}

The sun's position is not time itself; it is a physical state. A sundial maps this to a time reading. Similarly, $\mathcal{C}_{\text{trunk}}$ is a causal order, and τ_{branch} maps this order to a proper time on the branch. Both can be calculated indirectly, differing only in the length of the reconstruction chain.

Symbol Conventions (Unified Throughout, Four-Quantity Comparison Table):

Symbol	Operational Definition	Dimension / Value	Measurability	Note
$\Gamma_{\text{trunk,local}} \equiv \Gamma_{\text{trunk}}$	Local geometric dilation factor relative to ideal curvature-free trunk limit, $\Gamma_{\text{trunk,local}} = d\mathcal{C}_{\text{trunk}}/d\tau_{\text{branch}} = \sec\theta \geq 1$	Dimensionless, always ≥ 1	Indirect, converted from Γ_{local}	Larger Γ, local clock slower relative to ideal trunk
Γ_{local}	Operational measurement ratio relative to initial	Dimensionless, can be >, =, <	**Directly measured by**	Directly obtained by Layer-1

Symbol	Operational Definition	Dimension / Value	Measurability	Note
	handshake reference, $\Gamma_{\text{local}} = \Delta\tau_{\text{local}}/\Delta\tau_{\text{base}} = \Gamma_{\text{trunk,base}}/\Gamma_{\text{trunk,local}}$	1	**DPSP**	triangulation
$\gamma_{B/A}$	Relative time-rate between two physical endpoints, $\gamma_{B/A} = d\tau_B/d\tau_A$	Dimensionless	Indirect	Endpoint-to-endpoint, different from Γ_{local} relative to t_0 reference, must not be conflated
$\lambda(t)$	Target-to-source flow-rate ratio, $\lambda(t) = d\tau_{\text{target}}/d\tau_{\text{source}}$	Dimensionless	Indirect	Used for Layer-3 adaptive pacing

These four must not be conflated. When conversion first appears, remind in parentheses: $\Gamma_{\text{local}} = \Gamma_{\text{trunk,base}}/\Gamma_{\text{trunk,local}}$ (who divides whom). The protocol directly measures interval ratio Γ_{local}; causal order $\mathcal{C}_{\text{trunk}}$ is post-calculated integral reconstruction $\mathcal{C}_{\text{trunk}}(\tau_0) + \Gamma_{\text{trunk,base}} \int 1/\Gamma_{\text{local}} d\tau'$.

3. Historical Confusion of Absolute and Relative Time: TCLM's Decoupling Perspective

In physics history, Newtonian absolute time and Einsteinian relative time have long been viewed as incompatible.

Newtonian absolute time presupposes:

> A universal time background flowing uniformly, independent of matter, shared across the entire universe.

Einsteinian relative time points out:

> Time is not an independent background; it and space together constitute a geometric manifold. Under different gravitational potentials and motion states, time-flow rates inherently differ.

If the two are treated as mutually exclusive answers to the same question, conflict is inevitable.

However, TCLM points out that this conflict arises from conflating two different dimensions of description into one.

What Newton truly wanted to guarantee was not "identical time," but "irreversible causal order."
What Einstein truly described was not "reversible causality," but "local time flow varying with geometry."

In TCLM:

- Newton's absolute time → corresponds to the global causal order $\mathcal{C}_{\text{trunk}}$ and its mapping requirement $\tau_{\text{branch}}(\mathcal{C}_{\text{trunk}})$.
- Einstein's relative time → corresponds to the geometric branch latency $\Delta t_{\text{latency}}$.

They do not contradict, because they answer fundamentally different questions.

What Newton saw was the trunk of the tree; what Einstein saw was the branches.
They merely stood before the same tree, pointing out different dimensions.

Therefore, TCLM does not advocate returning to Newton, nor does it deny Einstein.
It merely places "causality" and "time" — which have been conflated — back into their respective positions.

Why the Lorentz Transformation's $t = 0$ Is Not Absolute:

The Lorentz transformation is correct when $v < c$. But its $t = 0$ relies on the Einstein synchronization convention, which is based on the speed of light. Different observers using light to synchronize clocks obtain different simultaneity slices, so $t = 0$ is not absolute. TCLM does not deny this result; TCLM merely points out that the non-absoluteness of $t = 0$ is a property of geometric projection and does not conflict with the absoluteness of the underlying causal order $\mathcal{C}_{\text{trunk}}$.

Chapter 3: Deep-Space Engineering Implementation: Dual-Pulse Synchronization Protocol (DPSP) and Initial-Handshake Causal-Order Reconstruction

Symbol Note: This chapter follows the "Symbol Conventions (Unified Throughout)" of Chapter 2. Unless otherwise subscripted, Γ refers to Γ_{local}. Γ_{trunk}, Γ_{local}, $\gamma_{B/A}$, and $\lambda(t)$ must not be conflated.

0. Operational Core

This section first explains the core operational principle of DPSP, before entering into terminology and formulas.

Core Principle: Measuring local clock interval ratios, rather than absolute time readings.

This operation does not involve one-way light-speed measurement, nor does it depend on the Einstein synchronization convention. What DPSP compares is the adjacent arrival intervals recorded by the same local clock, not the reading alignment of clocks at the two ends.

The specific operations are as follows:

1. The central node emits classical optical pulses at fixed local-clock intervals while simultaneously triggering a set of local reference events (currently envisioned as quantum-entanglement measurement events). The central node records its own emission timestamps and local reference-event timestamps.
2. The receiving ends record the local arrival timestamps of both the classical light and the local reference events upon receipt.
3. The receiving ends transmit these two timestamps back to the central node via a return channel. The return-channel timestamp is not included in the computation.
4. The central node uniformly receives the timestamps from all receiving ends, compares them with its own recorded emission and local reference-event timestamps, and computes the interval ratio Γ_{local}.

This operation has three important properties:

- **No prior clock synchronization between the two ends is needed.** The absolute readings of the emitter and receiver are irrelevant; what matters is the interval ratio.

- **The fluctuation of the emitter's clock is itself the signal.** If the emitter's clock runs fast, the emission interval shortens; if slow, it lengthens. This variation is directly reflected in the arrival interval measured at the receiving end. We do not need to know the theoretical "standard rate" of the emitter — only reconstruct its relative rate variation from the arrival intervals.
- **The main spatial perturbation source is the geometric variation of the propagation path and medium.** If the signal propagation path or medium dynamics changes between two samples, the measured interval at the receiving end will superpose "clock-rate difference" and "propagation-latency variation." The purpose of multi-baseline vector triangulation is to decouple these two types of variables.

Role of Multiple Baselines:

If the optical path is stable, a single baseline suffices. Multiple baselines are not for clock synchronization but for separating, when the optical path changes, "the geometric-latency variation proportional to baseline length" from "the clock-rate difference independent of baseline length."

Role of Quantum-Entanglement Event Timestamps (Optional Second Local Reference):

It is currently envisioned as the second set of local event timestamps, used to provide a local reference independent of the classical optical path. **This is not a necessary condition for the protocol, nor is it the most central part.** The core operation is always: the emitter emits at fixed local intervals; the central node computes the interval ratio.

This protocol does not rely on interpretations of "whether entanglement is theoretically synchronous," nor does it assume that the two measurement events have objective simultaneity. We only use the operable measurement results: both ends independently measure, and use local atomic clocks to record event timestamps, then ex-post compare via classical channel to compute the delay difference $\Delta\tau_{\mathrm{delay}} = t_{\mathrm{light}} - t_{\mathrm{ref}}$ (where t_{ref} is the local reference-event timestamp, whose operational requirements are the same as those envisioned for quantum entanglement:

1. Decoupled from spatial propagation path: its intrinsic latency $\Delta t_{\mathrm{intrinsic}}$ does not include the geometric path term $\int \sqrt{g_{ij}dx^i dx^j}/v_{\mathrm{signal}}$, independent of baseline length L_X
2. Sufficiently stable: its intrinsic jitter $\delta(\Delta t_{\mathrm{intrinsic}})$ is significantly smaller than the geometric delay difference to be measured $\Delta t_{\mathrm{prop}}(L_B) - \Delta t_{\mathrm{prop}}(L_A)$
3. Significantly smaller than optical latency: $|\Delta t_{\mathrm{intrinsic}}| \ll |\Delta t_{\mathrm{prop}}|$, can serve as a stable anchor

Quantum-entanglement measurement events are one candidate source for such events; a second independent atomic-ensemble spontaneous-transition trigger event, a locally independent ultra-stable frequency-standard self-trigger event, and other events with the same operational characteristics are equivalent candidates. DPSP only uses their operable delay difference $\Delta\tau_{\mathrm{delay}} = t_{\mathrm{light}} - t_{\mathrm{ref}}$, not depending on whether their physical mechanism is quantum entanglement.

The decoupling and stability described in this section are empirical premises, pending further experimental testing. If future experiments confirm that the effective latency of the chosen candidate has a non-negligible coupling to the local metric tensor $g_{\mu\nu}$, causing f_{light} and f_{ref} to be linearly dependent, then local spacetime triangulation degenerates and cannot break the geometric rotational degeneracy; in that case, it can be replaced by another candidate of the same class satisfying the above three conditions, and the protocol architecture remains unchanged.

Difference from Traditional Architectures:

Traditional deep-space timekeeping requires high-frequency bidirectional handshakes because it attempts to align the clock readings of both ends. DPSP does not require reading alignment — only the interval ratio. The initial handshake locks only one integration constant t_0, not clock readings.

The following sections are the engineering formalization, precision refinement, and error handling of this operational core.

0.1 Engineering Motivation & Necessity

Traditional deep-space timekeeping architectures assume the time-flow rate to be a predictable constant or static offset. Under the TCLM framework, however, the local time-flow-rate factor relative to the initial handshake (Γ_{local}) is intrinsically a continuously floating field affected by gravitational and velocity fields. This property gives rise to three major deep-space timekeeping bottlenecks, which also constitute the engineering necessity for this protocol:

1. **Local Imperceptibility**: When a single observation node is located on a fixed celestial body (e.g., Earth's surface), the local curvature-gradient variation is extremely small, so the time-flow-rate fluctuation is virtually imperceptible in short-term measurements. Observers easily mistake the time-flow rate for a constant, thereby overlooking the dynamic corrections needed for alignment relative to the initial handshake.

2. **Interplanetary Scale Amplification**: Once the endpoint crosses to deep space (e.g., the Earth–Mars system), the tiny differences in local curvature and potential are continuously accumulated and amplified through time integration. The Earth–Mars time-flow-rate difference can reach the $477\ \mu\text{s}/\text{day}$ order (with a $\pm 226\ \mu\text{s}/\text{day}$ variation depending on orbital position), and this difference varies dynamically with orbital eccentricity, solar gravitational-potential perturbations, and celestial perturbations, making long-term extrapolation with a single fixed constant impossible.

3. **Infeasibility of Continuous Sync**: If one attempts to rely on high-frequency bidirectional signals for continuous phase-locking, two limits arise: first, the **light-speed propagation-latency limit** (e.g., Earth–Mars one-way delay of 3 to 22 minutes, making real-time feedback impossible); second, the **unpredictability of continuous dynamic perturbations**, which causes any fixed mathematical extrapolation to diverge over time.

0.1.1 Applicability Boundary

DPSP is not intended to replace all deep-space timekeeping architectures, nor is it necessary in all scenarios. Its applicable conditions are:

1. **Sufficient local compute**: The node must be capable of cross-correlation, geometric reconstruction, and continuous integration. If local compute is insufficient, traditional telemetry-based timekeeping suffices.
2. **Tolerance of communication blackouts**: If the mission permits continuous high-frequency handshakes and the link is stable, traditional DSN timekeeping already meets requirements. The primary value of this protocol lies in providing offline autonomous timekeeping capability.
3. **Interplanetary or longer-term missions**: For near-Earth or short-term missions, the relative time-flow-rate difference is extremely small, and the engineering benefit of introducing this protocol is limited.

In short, DPSP is an offline time-reconstruction solution designed for "long-term, offline, high-compute" deep-space nodes, not a universally required solution.

0.2 Paradigm Shift: From "Earth-Model Dependency" to "Local Physical Measurement"

The fundamental limitation of traditional deep-space timekeeping (e.g., Deep Space Network) lies not only in the master-slave ideology of treating Earth as a privileged reference frame, but also in its methodological dependence on Earth-side astronomical formulas and ephemeris models. A deep-space node lacks the capacity to measure its own time-flow rate and can only passively receive calibration parameters computed on Earth. Once communication is interrupted or the ephemeris model becomes invalid, the node loses temporal autonomy.

Under the TCLM framework, Earth time (t_{Earth}) and Mars time (t_{Mars}) are both merely "geometric branch residues" under local gravity wells and motion states; neither is an objective time reference of the universe.

The fundamental breakthrough of DPSP lies in granting any deep-space node the ability to perform "local physical measurement." The node no longer needs to query Earth's gravitational-field formulas; it directly measures its own time-flow-rate variation relative to the initial handshake (Γ_{local}) via local event interval-ratio measurement. Earth-computed models serve only as initial references, not as the sole source of truth.

Conceptual Dimension	Traditional Deep-Space Timekeeping (DSN-type)	Dual-Pulse Synchronization Protocol (DPSP / TCLM Topological)
Reference-frame positioning	Earth-privileged reference frame (Master-Slave)	Decentralized, no-privileged node (Decentralized Nodes)
Alignment target	Follow Earth time t_{Earth}	Directly measure time-flow-rate variation relative to the initial handshake (Γ_{local})
Communication dependency	High-frequency/continuous bidirectional handshake to maintain phase lock	Only initial-value lock; thereafter, local differential autonomous computation
Physical essence	Engineering compromise to compensate for signal delay	Observer's decoding of the universe's underlying topological causal order

Based on the above paradigm shift, this protocol abandons the traditional high-frequency bidirectional phase-locking architecture, decoupling the "local time-flow-rate factor (Γ_{local})" and the "global initial constant (t_0)" at a dual-logical level, and establishing a multi-layer dynamic control system. This mechanism may be termed **Decentralized Initial-Constant Locking and Local Rate Measurement**.

This paper uses the Earth–Mars system as the first-stage engineering calibration case. The applicability of DPSP is not limited to interplanetary distances; provided an initial handshake can be completed, the subsequent autonomous reconstruction and communication-blackout tolerance can, in principle, be extended to interstellar and even intergalactic scales.

DPSP's Output Is Causal Order; Time Is Derived from Causal Order.

It should be particularly emphasized: the objective of this protocol is not to align the clock readings of the two ends, nor to directly measure "time." Under the TCLM framework, the causal order $\mathcal{C}_{\mathrm{trunk}}$ is fundamental, and

the time reading $\tau_{\text{branch}}(\mathcal{C}_{\text{trunk}})$ is a product mapped from the causal order. This relationship is like the sundial: the sun's position in the sky is not time itself; the sundial maps this physical state into a time reading. Similarly, the causal order is not time; τ_{branch} maps the causal order to the branch proper time.

What DPSP reconstructs is the causal order relative to the initial handshake:

$$\mathcal{C}_{\text{trunk_reconstructed}}(\tau) = t_0 + \int_{\tau_0}^{\tau} \Gamma_{\text{trunk,local}}(\tau')d\tau' = t_0 + \Gamma_{\text{trunk,base}} \int_{\tau_0}^{\tau} \frac{1}{\Gamma_{\text{local}}(\tau')} d\tau'$$

When the base is normalized to $\Gamma_{\text{trunk,base}} = 1$, this simplifies to $\mathcal{C}_{\text{trunk}} = t_0 + \int_{\tau_0}^{\tau} \frac{1}{\Gamma_{\text{local}}(\tau')} d\tau'$.

Once the causal order is reconstructed, the local time reading can be obtained by mapping $\tau_{\text{branch}}(\mathcal{C}_{\text{trunk}})$. The absolute-reading offset between the two clocks naturally cancels in this process and does not affect the reconstruction of the causal order. What the initial handshake locks is the integration constant t_0, not clock alignment.

---

1. Layer 1: Local Spacetime-Vector Triangulation

Node Hardware Configuration

The basic node configuration comprises:

1. **Local physical time-frequency reference**: high-precision optical lattice clock or atomic clock, providing constant proper frequency f_0 and recording local proper time τ;
2. **Local reference event source**: providing local reference-event timestamp t_{ref} decoupled from spatial propagation path (must satisfy three operational conditions: decoupled from spatial path, sufficiently stable, significantly smaller than optical latency; quantum-entanglement source is one candidate);
3. **Dual/Multi optical baseline transceiver modules**: at least two optical baselines of different lengths with coincident origins, recording $t_{\text{light},A}$, $t_{\text{light},B}$;
4. **Multi-baseline cross-correlation processing unit**: performing Layer-1 local solving of Γ_{local}.

Architecture Decoupling: Logging and Rendering

DPSP consists of two layers; the flow clock depends only on Layer 1:

1. **Layer 1 (Local):** Node locally records $t_{\text{light},A}$, $t_{\text{light},B}$ and t_{ref} per f_0, where t_{ref} serves as absolute origin of local trigger phase (Epoch Anchor) to eliminate internal circuit pulse-origin uncertainty of $t_{\text{light},A}$ and $t_{\text{light},B}$, locally paired to obtain $\Delta D = (L_B - L_A) \cdot \eta_{\text{eff}}$, solving local time flow rate Γ_{local}. This layer requires no cross-node handshake.

2. **Flow Clock (FASC):** Independent display unit after Layer 1, not intervening in DPSP closed loop. It receives locally solved Γ_{local}, integrates and synthesizes $\mathcal{C}_{\text{trunk}} = \int \Gamma_{\text{trunk,local}} d\tau + \mathcal{C}_0$ and outputs. Earth node and Mars node each solve locally and output independently; their flow clocks align without continuous handshake. $\mathcal{C}_0$ is negotiated only once at initial link establishment as Root Session ID. If a local node experiences restart or long-term out-of-sync, only a Layer-2 lightweight handshake is triggered to recalibrate $\mathcal{C}_0$, without interrupting the intrinsic integration of Γ_{local} in Layer 1.

3. **Layer 2 (Cross-node):** Used only to solve geometric deflection angle θ and medium corrections such as Shapiro delay, requiring cross-node classical channel handshake, and does not affect local ticking of the flow clock.

> **Operational Definition (Principle Level):** Earth time (absolute simultaneity) is unattainable; Earth flow-clock time (causal progress $\mathcal{C}_{\text{trunk}}$) is locally obtainable, Mars flow-clock is also locally obtainable, both pointing to the same trunk coordinate.

Definition of Dual-Pulse

The "dual-pulse" referred to in this protocol means that at least two sets of baselines of different lengths are established at the local end from the same master clock, with two independent pulse units triggered simultaneously. Its structure is differential measurement:

```
Master Clock
├─ [A short L_A] light A1 — light A2 / ref A1 — ref A2
└─ [B long L_B] light B1 ——— light B2 / ref B1 ——— ref B2  (L_B > L_A)
```

Each pulse unit contains:

1. Classical optical timestamp t_{light}: affected by geometric distance L
2. Local reference-event timestamp t_{ref}: currently envisioned as quantum-entanglement measurement event, not traveling through geometric path, stable anchor

A single baseline provides only a one-dimensional projection of the delay difference and cannot uniquely solve for the geometric deflection angle of the spacetime manifold. Therefore, the minimum requirement of the dual-pulse is two baselines of different lengths; if a third baseline of different length is available, a closure method can further verify the uniqueness and measurement stability of the θ solution.

A single baseline yields only one clock-difference $D_X = t_{\text{light},X} - t_{\text{ref},X}$, unable to separate rate and bias, hence three-tier layered definition:

- **Minimum operation (1D rate, DPSP core)**: 2 collinear, different-length baselines ($L_B \neq L_A$) suffice.
- **Full geometry (2D deflection)**: requires 2 non-collinear, different-length baselines.
- **Engineering verification (closure)**: 3 or more different-length baselines, verifying uniqueness of θ.

1.1 Geometric Insufficiency of a Single Baseline

Local rigidity assumption: L_A, L_B refer to intrinsic lengths measured in the node's local proper-time frame, assuming baseline scale $L \ll$ curvature radius, treated as locally rigid; $L_B - L_A$ is pre-measured by local calibration, its second-order variation $O(L^2 R)$ with g_{ij} neglected.

A single baseline provides only one scalar clock-difference (unit s):

$$D_A = t_{\text{light},A} - t_{\text{ref},A} = \frac{L_A}{c} \cdot f(\Gamma) + bias = L_A \cdot \eta_{\text{eff}} + bias$$

where L_A has unit m, c has unit m/s, $f(\Gamma)$ is dimensionless correction function, $\eta_{\text{eff}} \equiv f(\Gamma)/c$ has unit s/m, $bias$ has unit s.

After introducing the second different-length baseline L_B:

$$D_B = t_{\text{light},B} - t_{\text{ref},B} = \frac{L_B}{c} \cdot f(\Gamma) + bias$$

$$\Delta D = D_B - D_A = \frac{L_B - L_A}{c} \cdot f(\Gamma) = (L_B - L_A) \cdot \eta_{\text{eff}}$$

The bias is eliminated by differencing, $(L_B - L_A)$ provides radial decomposition, uniquely solving $\Gamma_{\text{trunk}} = \sec\theta$. This is the principle by which DPSP solves the rate via interval ratio, then synthesizes $f_{\text{trunk}} = \Gamma_{\text{trunk,local}} \cdot f_0$ via rate-modulation to form the flow clock.

1.2 Functional Positioning of the Single Composite Pulse and Degeneracy Removal

A traditional one-way classical optical signal provides only a scalar delay, which cannot distinguish whether the delay arises from spatial distance or time-flow-rate tilt.

The single pulse unit of this protocol is composed of a quantum-entanglement event timestamp and a classical optical return timestamp:

1. Quantum-entanglement event timestamp: recorded by the local atomic clock, unaffected by the classical propagation-path delay.
2. Classical optical return timestamp: recorded by the local atomic clock, affected by spatial distance and local curvature.

After ex-post comparison, the two can partially separate spatial propagation delay from time-flow-rate-related delay.

This function does not depend on any specific interpretation of quantum mechanics, nor does it claim that quantum entanglement can directly observe the trunk.

Under the TCLM framework, the quantum-entanglement event timestamp can at most be regarded as a second set of local measurements with a different delay-response mechanism from the classical optical timestamp, not the trunk itself. Operational definition: the entanglement event in DPSP carries no remote information transmission, serving only as a "local event trigger" for the local endpoint upon receiving the particle. Its core value is that the occurrence mechanism of this event does not go through the optical propagation process in geometric space, thus operationally providing a local timestamp decoupled from the spatial propagation path. This correspondence is an operational hypothesis within the framework, relying on the premise that "entanglement-correlation latency is below current measurement limits," and does not constitute an ontological claim of physical fact.

Key Physical Conditions and Limitations

There are three core conditions for DPSP's local decoupling mechanism to hold:

First, entanglement latency and optical latency must be affected by different factors. Optical latency is affected by spatial path, propagation speed, and local curvature; what affects entanglement latency is not yet fully understood. Based on existing Bell-test experiments (e.g., Aspect 1982, Zeilinger 1998, etc.), "local hidden-variable mechanisms at or below the speed of light" have been ruled out; its mechanism is not the same as optical latency. If the two are affected by exactly the same factors, they will cancel upon subtraction and cannot be separated. If they are affected by different factors, they will not cancel upon subtraction, and partial spatial delay and time-flow-rate-related delay can be separated.

Second, entanglement latency must be sufficiently stable. If its fluctuation is too large, then after subtraction only noise remains, and no effective signal can be extracted.

Third, entanglement latency must be significantly smaller than optical latency, to serve as a stable anchor. At deep-space scales, optical latency is on the order of minutes, while entanglement latency is indistinguishable under current precision; this condition is naturally satisfied.

Under current optical-lattice-clock precision, the effective latency of entanglement correlation is too small to be distinguished. This is not "assuming it is synchronized," but "current precision is insufficient to detect its difference from zero." However, whether this latency will cause observable error over long-term accumulation remains to be tested experimentally. If future experiments find that this correlation has a measurable response to local gravitational potential or velocity field, then other local reference events unrelated to spatial propagation path must be used.

This protocol does not need to separate the specific values of optical latency and entanglement latency; what is measured is the difference between the two timestamps recorded by the local clock for the same set of events. This difference is a combination of the two. As long as the two are affected by different factors, this combination can be decomposed into components distinguishable from optical latency.

1.3 Operational Independence of DPSP

DPSP's local measurement architecture does not depend on high-frequency bidirectional phase-locking, nor does it depend on the definition of absolute simultaneity. However, the second-layer initial handshake still depends on a single ephemeris and a one-way light-speed convention; this is the engineering premise for the one-time alignment constant t_0.

Its operational independence comes from two sources:

- Optical event timestamp: affected by propagation delay.
- Entanglement event timestamp: a local record, unaffected by geometric-path delay; but its pairing relationship with the optical event still needs to be confirmed ex-post via a classical channel, without involving faster-than-light communication.

Unless otherwise specified below, $\Delta t_{\mathrm{latency}}$ adopts the local clock-difference definition, i.e.:

$$\Delta t_{\mathrm{latency}} \equiv \Delta \tau_{\mathrm{delay}} = t_{\mathrm{light}} - t_{\mathrm{ref}}$$

> **Same-clock measurement note**: t_{light} and t_{ref} are both proper-time differences measured by the same local atomic clock, hence $\Delta \tau_{\mathrm{delay}} = \Delta \mathcal{C}_{\mathrm{delay}} / \Gamma_{\mathrm{trunk,local}}$. If future experiments confirm that the chosen t_{ref} candidate has a non-negligible coupling to $g_{\mu\nu}$, causing f_{light} and f_{ref} to be linearly dependent, then triangulation degenerates; in that case it can be replaced by another candidate of the same class satisfying the above three conditions, and the protocol architecture remains unchanged.

where both t_{light} and t_{ref} are local atomic-clock readings, t_{ref} is the local reference-event timestamp (currently envisioned as quantum-entanglement measurement event). Layer-1 triangulation solves Γ_{trunk}, which is then converted to $\Gamma_{\mathrm{local}} = \Gamma_{\mathrm{trunk,base}} / \Gamma_{\mathrm{trunk,local}}$.

If the local clock difference $\Delta \tau_{\mathrm{local}}$ needs to be converted to the initial-handshake reference, then under reference normalization:

$$\Delta t_{\mathrm{latency}}^{(\mathrm{base})} = \frac{\Delta \tau_{\mathrm{local}}}{\Gamma_{\mathrm{local}}}$$

Unless otherwise superscripted, $\Delta t_{\mathrm{latency}}$ always refers to the local clock-difference version.

The initial value t_0 is a conventional integration constant; all subsequent quantities are causal-order reconstructions relative to t_0.

This independence relies on the operational premise described in Section 1.2; this premise is not a necessary condition for DPSP to hold, but its precision remains to be further tested experimentally.

1.4 Topological Time Synthesis and Causal Precedence Norm

DPSP's operational core is: to obtain the global causal rate $\Gamma_{\text{trunk,local}}$ via local measurement, then synthesize the trunk coordinate $\mathcal{C}_{\text{trunk}}$ via the Flow-Adaptive Synthesized Clock (FASC).

1. Physical Mechanism: Flow-Adaptive Synthesized Clock (FASC)

FASC adopts a dual-layer architecture of "physical proper layer + digital rate-modulation layer":

- **Physical Layer**: Local high-precision time-frequency reference (optical lattice clock or atomic clock) provides constant proper frequency f_0 and proper time τ. This layer is never steered, continuously accumulating twin-paradox proper-time differences.
- **Processing Layer (DSP)**: DPSP computes in real time the geometric dilation factor $\Gamma_{\text{trunk,local}} \equiv \frac{d\mathcal{C}_{\text{trunk}}}{d\tau} = \sec\theta \geq 1$ and its operational ratio relative to the handshake base $\Gamma_{\text{local}} \equiv \Delta\tau_{\text{local}}/\Delta\tau_{\text{base}} = \Gamma_{\text{trunk,base}}/\Gamma_{\text{trunk,local}}$ (can be greater than, less than, or equal to 1), and modulates via NCO:

$$f_{\text{trunk}}(\tau) = \frac{\Gamma_{\text{trunk,base}}}{\Gamma_{\text{local}}(\tau)} \cdot f_0 = \Gamma_{\text{trunk,local}}(\tau) \cdot f_0$$

- **Dynamic Compensation Logic**: When $\Gamma_{\text{local}} < 1$ (local proper time runs slow relative to base, e.g., deep gravity well / high speed, large $\Gamma_{\text{trunk,local}}$), $f_{\text{trunk}} = \Gamma_{\text{trunk,base}}/\Gamma_{\text{local}} \cdot f_0$ is automatically increased; when $\Gamma_{\text{local}} > 1$ (local runs fast, e.g., high gravitational potential, small $\Gamma_{\text{trunk,local}}$), f_{trunk} is decreased. This keeps FASC outputs $\mathcal{C}_{\text{trunk}}$ advancing consistently relative to t_0, not aligning to a fictitious global trunk clock.
- **Trunk Coordinate Synthesis**: Total Tick output by FASC is real-time $\mathcal{C}_{\text{trunk}}$:

$$\mathcal{C}_{\text{trunk}}(\tau) = \mathcal{C}_{\text{trunk}}(\tau_0) + \Gamma_{\text{trunk,base}} \int_{\tau_0}^{\tau} \frac{1}{\Gamma_{\text{local}}(\tau')} d\tau' = \mathcal{C}_{\text{trunk}}(\tau_0) + \int_{\tau_0}^{\tau} \Gamma_{\text{trunk,local}}(\tau') d\tau'$$

Nodes need not know each other's UTC; only need to align scalar $\mathcal{C}_{\text{trunk}}(\tau_0) = t_0$ via initial handshake.
Engineering Alignment Criterion: After initial t_0 locking, the two ends' FASC outputs $\mathcal{C}_{\text{trunk}}$ satisfy $|\mathcal{C}_{\text{trunk,Earth}} - \mathcal{C}_{\text{trunk,Mars}}| < \mathcal{O}(10^{-9}\ \text{s})$, at which point the flow clocks are said to be aligned (budget target, pending end-to-end validation, not achieved precision).

2. DPSP Causal Precedence Rule

All determinations directly read FASC-synthesized $\mathcal{C}_{\text{trunk}}$:

1. **Single node**: $A \prec B \iff \mathcal{C}_{\text{trunk}}(A) < \mathcal{C}_{\text{trunk}}(B)$
2. **Cross-node**: Premise: both nodes have solved geometric delay $\Delta t_{\text{latency}}$ via HDP (containing $\mathcal{C}_{\text{trunk,send}}$, Γ_{local}, $\Delta\tau_{\text{pulse}}$) and normalized FASC $\mathcal{C}_{\text{trunk}}$ to same t_0. If not locked, cross-node order is undecidable.

3. **Indistinguishable zone**: If $|\mathcal{C}_{\text{trunk}}(A) - \mathcal{C}_{\text{trunk}}(B)| < \sigma_\Gamma \cdot \Delta\tau$, indistinguishable within precision. This does not constitute privileged simultaneity, nor changes $\text{sgn}(\Delta\mathcal{C}_{\text{trunk}}) \equiv +1$.

> **Core**: Observed arrival time $t_{\text{observed}} = \tau_{\text{branch}} + \Delta t_{\text{latency}}$ is only geometric latency, never used for precedence. Sole basis is FASC $\mathcal{C}_{\text{trunk}}$.

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2. Layer 2: Interplanetary Initial-Handshake Pulse (Deep-Space Initial-Constant Locking)

Because the global causal-order reconstruction equation necessarily contains an unknown **initial integration constant (t_0)**, solving the local $\Gamma(t)$ variation alone yields only the time-advance quantity, not the initial alignment constant between the two time axes.

- **Single-handshake procedure**:
    1. The reference end (e.g., Earth) emits a single classical deep-space radio pulse carrying the initial timestamp $t_{\text{Earth_start}}$ to the target end (e.g., Mars).
    2. The Mars end receives this pulse and, combining orbital geometric distance and multi-frequency dispersion compensation, computes the light-speed delay $\Delta t_{\text{propagation}}$, thereby establishing the initial alignment relationship between Mars local time and Earth reference time (t_0).
- **Autonomous-computation state**: Once the initial value t_0 is locked, the Mars end can cease high-frequency deep-space communication and rely purely on the $\Gamma_{\text{local,Mars}}(\tau)$ obtained in real time from "Layer-1 local triangulation" to independently perform causal-order autonomous reconstruction relative to the initial handshake:

$$\mathcal{C}_{\text{trunk_reconstructed}}(\tau) = t_0 + \Gamma_{\text{trunk,base}} \int_{\tau_0}^{\tau} \frac{1}{\Gamma_{\text{local,Mars}}(\tau')} \, d\tau'$$

  where $\Gamma_{\text{trunk,base}}$ is locked by the initial handshake. If the reference node is defined as curvature-free and normalized to $\Gamma_{\text{trunk,base}} = 1$, this reduces to $\mathcal{C}_{\text{trunk_reconstructed}}(\tau) = t_0 + \int_{\tau_0}^{\tau} \frac{1}{\Gamma_{\text{local,Mars}}(\tau')} \, d\tau'$.
- **Blackout re-synchronization mechanism**: If a deep-space node enters a long-term communication blackout due to solar occultation, the local continuous-field integration engine can still independently track time-flow-rate variation without causing a causal-order discontinuity. However, the t_0 locked by the initial handshake is a one-time classical alignment parameter; it does not drift due to local integration. The purpose of re-handshaking after blackout is to re-confirm the alignment relationship between the two time axes and to correct engineering errors possibly accumulated during local integration. The geometric curvature measurement of the time-flow rate itself does not fail due to communication interruption.

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3. Layer 3: Continuous Rate Dynamic Control (Adaptive Pacing, Quadratic Compensation, and Causal Backpressure)

To address the frequency asymmetry and data-queue overflow problems arising from the continuously fluctuating time-flow rate ($\frac{d\Gamma}{dt} \neq 0$), the protocol introduces the following dynamic control mechanisms:

- **Adaptive Pulse Frequency Scaling**: The two ends exchange their respective local time-flow-rate curves $\Gamma_{\text{local}}(t)$ during the initial handshake. The local end can then combine its own $\Gamma_{\text{local,source}}(t)$ with the

received $\Gamma_{\text{local,target}}(t)$ to reconstruct the relative flow-rate ratio $\lambda(t) = \frac{d\tau_{\text{target}}}{d\tau_{\text{source}}}$. The high-rate end (source) automatically adjusts its timekeeping pulse emission period:

$$\Delta\tau_{\text{source}} = \frac{\Delta\tau_{\text{target}}}{\lambda(t)}$$

When the target end is in a low-rate region ($\lambda \ll 1$), the source automatically lengthens its emission interval to match the target's local processing frame rate, avoiding high-frequency pulse bombardment of the low-rate end.

- **Continuous Curvature Quadratic Compensation**: In a continuously varying gravitational or acceleration field, the dual-pulse signal header is forced to carry the node's current "local time-rate acceleration and gravitational-gradient derivative ($\frac{d\Gamma_{\text{local}}}{d\tau}$)." The receiving end uses a Taylor-expansion quadratic term for continuous curvature fitting:

$$\Gamma_{\text{local}}(\tau + \Delta\tau) \approx \Gamma_{\text{local}}(\tau) + \frac{d\Gamma_{\text{local}}}{d\tau}\Delta\tau + \frac{1}{2}\frac{d^2\Gamma_{\text{local}}}{d\tau^2}(\Delta\tau)^2$$

  The receiving end can thus locally predict the clock drift between two discrete pulses without relying on high-frequency pulses, resolving the accumulated phase error caused by continuous acceleration fields.

  Phase perturbations from the solar wind and interplanetary plasma can be regarded as quasi-common-mode for all nodes within the solar system, but not strictly common-mode. Through multi-node cross-comparison and differential computation, some of these background perturbations can be partially suppressed without establishing precise individual models.

  Shannon-Nyquist Hard Boundary of Discrete Sampling: This compensation is intrinsically a discrete-sampling system. Let pulse sampling frequency be $f_{sample} = 1/T_{sample}$, environment field variation frequency be f_{field}. When $f_{field} > \frac{1}{2}f_{sample}$ (i.e., $T_{field} < 2T_{sample}$), e.g., $T_{sample} = 0.5s$ but large fluctuation within $0.1s$, aliasing and information loss must occur; quadratic Taylor expansion not only fails to correct but may misjudge high-frequency noise as low-frequency trend causing oscillation. This is a physical limit of all discrete systems, not a design flaw.

- **High-Rate Inertial Bridging & Adaptive Trigger**: DPSP is discrete sampling; there is a blind zone for $\Gamma_{\text{local}}(t)$ variation between measurements. To address the Nyquist limit, nodes may optionally be equipped with high-rate accelerometer (or IMU), whose role is strictly limited to kinematic high-frequency bridging and emergency trigger, not replacing dual-pulse geometric solving (per equivalence principle, IMU cannot alone distinguish gravity from kinematic acceleration):

  - Estimate kinematic components of $d\Gamma_{\text{local}}/d\tau$ and $d^2\Gamma_{\text{local}}/d\tau^2$, strengthening quadratic compensation
  - **Adaptive Trigger**: When IMU detects high-G drastic change (engine ignition, close-orbit maneuver), force DPSP to switch from low-frequency mode (e.g., $0.5s$) to high-frequency mode (e.g., $0.01s$) to cover the drastic period with high sampling
  - Consistency constraint with Γ_{local} sequence solved by dual-pulse; gravitational-potential-difference solving remains provided by clock interval ratio itself

  Model Boundary Statement: Flow-rate dynamic tracking capability is strictly bounded by Shannon information limit of discrete sampling. For ultra-high-frequency gravitational micro-perturbations or

drastic maneuvers with period smaller than pulse interval, system cannot resolve in real time and must rely on above high-frequency adaptive trigger and a priori gravity-field model.

- **Causal Backpressure and Queue Mechanism**: When the flow-rate ratio between source and target suddenly changes drastically (e.g., the target suddenly enters a strong gravitational well), causing the target's local causal-processing rate to lag behind the upstream input rate, the target's ingress queue triggers a critical backpressure signal.
    - The signal forces upstream nodes to reduce data-transmission throughput.
    - Overloaded event data is temporarily stored in the local node's "Causal Buffer," queued according to the monotonicity constraint $\frac{d\mathcal{C}_{\text{trunk}}}{d\tau} > 0$.
    - Messages are stretched for transmission in the delay pipeline; provided the buffer capacity is sufficient, buffer overflow or event loss is avoided for any low-rate sub-branch due to time dilation.

  This buffer is a local node engineering-queue mechanism whose capacity is constrained by the node hardware and local causal-processing rate. J^{cap} is a parameter in the theoretical model characterizing the local node's information-throughput limit; its precise correspondence to the actual engineering buffer is left to subsequent engineering teams.

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4. Theoretical Closure and Engineering-Boundary Decoupling

- **Theoretical Resolution Limit**: After removing medium and geometric perturbations, the theoretical resolution of this protocol under a geometric closed loop is constrained by the hardware limit of current optical-lattice clocks. In terms of fractional frequency stability, the current level has reached the 10^{-18} to 10^{-19} order of magnitude; when integrated over 1 day, this corresponds to an absolute time error of approximately 10^{-13} to 10^{-14} s. This is the hardware resolution limit, not a solo contribution of the protocol.

- **Deep-Space Engineering Error Budget**: In real deep-space, radio signals suffer solar-wind plasma dispersion, solar gravitational lensing, and ephemeris uncertainties (uncompensated uncertainty can reach $\sim 10^{-6}$ to 10^{-8} s), very likely drowning the tiny difference of Γ_{local} in environmental noise. Local optical lattice clock hardware resolution 10^{-18} to 10^{-19} (about 10^{-13} to 10^{-14} s per day) is only theoretical upper bound, not solo contribution of protocol. After multi-frequency dispersion compensation and ephemeris constraints, engineering alignment precision budget target is nanosecond-level ($\mathcal{O}(10^{-9}\ \text{s})$), this is a budget target pending end-to-end link simulation and empirical validation, not achieved precision. Gap of about 4-5 orders of magnitude between hardware limit and engineering target mainly comes from deep-space link SNR attenuation, plasma residual, and ephemeris uncertainty, constituting primary hard-core battlefield for physical deployment.

Chapter 4: Astrophysical-Measurement Data Calibration and Earth–Mars Transfer Example

(Note: The $\gamma_{B/A}$ used in this chapter is defined as the **relative** time-rate variation between two physical observation endpoints, and must be strictly distinguished from the local time-flow-rate factor Γ_{local} relative to the initial handshake in Chapter 3.)

1. Surface and GPS Satellite Regional Calibration Case (Earth Calibration)

Considering the combined special- and general-relativistic effects in low-Earth orbit (velocity slows by 7 μs/day, gravitational potential speeds by 45 μs/day), the measured net drift rate between the surface reference end and the GPS branch end is +38 μs/day. Substituting into the two-endpoint relative rate-variation formula:

$$\gamma_{\mathrm{GPS/Earth}} = \frac{d\tau_{\mathrm{GPS}}}{d\tau_{\mathrm{Earth}}} \approx 1 + 4.40 \times 10^{-10}$$

This data serves purely as a calibration example for "single-point rate variation can be measured and reconstructed" within a regional gravity well.

2. Earth–Mars Interplanetary Docking and Worldline Path-Integration Calibration

Let the Earth geoid and the Mars surface be independent physical branch endpoints. According to the complete calculation of NIST Ashby & Patla (2025), the Mars atomic clock averages about 477 μs per day faster than the Earth geoid, with a ±226 μs/day variation with orbital eccentricity.

This model uses continuous worldline field integration as the primary dynamic-alignment core. All time-flow rates are intrinsically continuously floating dynamic fields; the previously named "local static chain-transfer law (e.g., $\gamma_{C/A} = \gamma_{C/B} \cdot \gamma_{B/A}$)" is merely a discrete special case to which the integral operation naturally degenerates when the gravitational-potential perturbation tends to zero.

In relay communication, the satellite serves only as an "engineering signal-forwarding (edge-router)" function and does not constitute a necessary intermediary node for physical causal computation. The Mars end needs only to receive the single initial-handshake pulse from Earth; combined with the Layer-3 dynamic pacing and backpressure mechanisms, it can perform autonomous reconstruction via local triangulation and continuous integration. **The following budget takes a local 100 s continuous-field integration window as an example (not Earth-Mars two-way light travel 4-22 min), as an order-of-magnitude estimation of engineering alignment error, not achieved precision.**

Error Source	Uncompensated Magnitude	Compensated Residual	Note
Plasma dispersion	$\sim 10^{-6}$ s	$\sim 10^{-10}$ s	Residual after multi-frequency (S/X/Ka) $\propto 1/f^2$ fitting
Ephemeris uncertainty	$\sim 10^{-8}$ s	$\sim 10^{-9}$ s	After JPL ephemeris radial tens-of-meters constraint, dominant term
Clock hardware limit	10^{-18} to 10^{-19} (fractional stability)	10^{-16} s@100s	$10^{-18} \times 100$s, negligible within window
Integration-window accumulation	$\sigma_\Gamma \cdot \Delta\tau$	Depends on σ_Γ and window length	Integral drift of Γ_{local} measurement error
Signal-to-noise attenuation	Depends on link	Depends on link	Deep-space link R^{-2} attenuation

Total budget estimated by root-sum-square $\sigma_{\text{total}} = \sqrt{\sum \sigma_i^2}$, $\sqrt{(10^{-10})^2 + (10^{-9})^2} \approx \mathcal{O}(10^{-9}\ \text{s})$, hence engineering alignment error-budget target is nanosecond-level. This is a budget target pending end-to-end link simulation and empirical validation; theoretical limit is constrained by optical lattice clock hardware $10^{-18} \sim 10^{-19}$, not achieved by protocol alone.

Chapter 5: Future Extension Concepts (Unverified Capabilities): Gravitational-Wave Response Exploration Based on Dual-Pulse Timing Networks

1. **Spacetime-Perturbation Response (Conceptual)**: If a dual-pulse dynamic-curvature-solving endpoint network (e.g., an Earth–Mars timing network) is operated over the long term, its solved $\Gamma_{\text{local}}(t)$ curve may, in principle, not only reflect regular orbits and gravitational potential but also exhibit transient distortions in response to gravitational waves (metric perturbation, $h_{\mu\nu}$) sweeping through the local spacetime. This is a conceptual expectation, currently without a complete response function or sensitivity model.

2. **Chronometric Gravitational-Wave Sensing (Early Concept)**: A long-term multi-node $\Gamma_{\text{local}}(t)$ monitoring network may, under noise conditions and instrument-stability allowances, in principle respond to spacetime perturbations (including gravitational waves) sweeping through the region. This direction is worth exploring as one future possible application; however, no complete sensitivity, noise-spectrum, frequency-band, or integration-time budget currently exists. It is an early concept highly dependent on actual noise-suppression effectiveness, not a verified capability of this protocol.

This paper does not claim that this network already has gravitational-wave detection capability, nor does it claim that it can replace or surpass existing interferometers. This chapter is only a conceptual discussion of a future research direction.

Chapter 6: Theoretical and Engineering Pressure Tests and Boundary Defenses (Comprehensive Q&A)

No. & Question	Answer & Boundary Defense
Q1: How to explain the permanent time difference of the "twin paradox"?	In the TCLM framework, the two atomic clocks experience different spatial-motion trajectories (acceleration and gravitational fields); their accumulated branch curvature delays are inherently unequal. This is a curvature difference of geometric branches, not a causal reversal on the trunk.
Q2: If FTL holds, can signals be sent to the past or causal reversal be realized?	Under TCLM postulates, no by means of speed breakthrough. This paper only argues FTL is not a means to return to the past; physical limit of extreme speed is latency compression to zero: $\lim_{\eta \to 0^+} L_{\text{propagation}}(\eta) = 0$ (i.e., $L = 0$, Ping=0), not $L < 0$. Since $L_{\text{propagation}} \in [0, \infty)$ is non-negative closed interval, $L < 0$ undefined, thus no negative-latency path caused by speed exists. Time-machine via this path is closed in operational definition.
Q3: After matter falls into a black-hole event	Under TCLM's topological causal structure, not necessarily. The black-hole event horizon is only a geometric gateway where branch curvature reaches its limit; within the model, the evolution of matter on the causal trunk is still set

No. & Question	Answer & Boundary Defense
horizon, does time stop or flow backward?	to be strictly forward. The observable time near the event horizon and external-observer readings still require separate modeling and cannot be directly inferred from the causal order of the trunk alone.
Q4: Does the TCLM causal order break tensor covariance?	No. TCLM distinguishes "geometric observables" from "topological causal order." Local geometric observables (e.g., clock tick rate $d\tau$ and geometric latency $\Delta t_{\text{latency}}$) obey Lorentz covariance and the general-relativistic metric equation (as a local effective field theory). $\mathcal{C}_{\text{trunk}}$ serves only as a causal-sorting index of the global topological directed acyclic graph (DAG) and does not interfere with the metric-tensor operation of any local inertial frame.
Q5: Does the TCLM framework face the relativity limit of "mass-energy divergence near the speed of light"?	This paper does not claim to have resolved this limit. Type A (relative-distance FTL) and Type B (discrete FTL / topological transition, see Appendix A) are merely phenomenological classifications under the TCLM framework, used to explain "if some FTL mechanism exists, how the causal direction is maintained by the model." This paper does not claim that Type A or Type B already has engineering feasibility, nor does it claim that the mass-energy divergence limit has been bypassed.
Q6: Is TCLM a disguised revival of Newtonian "absolute spacetime" or the "ether"?	No. Newtonian absolute spacetime presupposes a "global absolute clock ticking synchronously at all points" and a rigid spatial medium. TCLM's $\mathcal{C}_{\text{trunk}}$ is not a global synchronized clock but a second-parameterized index of causal chronological order. Its strict monotonic increase guarantees only the causal direction; it does not provide an observable privileged simultaneity slice. The chronological order of $\mathcal{C}_{\text{trunk}}$ is only a mathematical index of causal order; it does not constitute a physically observable simultaneity plane. TCLM also does not introduce a mechanical ether or a preferred reference frame. Under the TCLM framework, all physically measured time (clock time) is a geometric path integral on the branch manifold.
Q7: Does the non-local correlation of quantum-entanglement measurement violate the no-communication theorem or cause trunk disorder? (Its non-locality has been confirmed by Bell tests; see Chapter 3, Section 1.2.)	No. In this protocol, quantum entanglement serves only as one local measurement event within each pulse unit, recorded by the local atomic clock as an event timestamp. This timestamp is compared ex-post with the classical optical return timestamp. The dual-pulse itself relies on two or more independent baselines, rather than using quantum entanglement for real-time communication. Single-sided measurement yields only random eigenvalues and cannot immediately read the opposite end's time; the system uses ex-post comparison of the two timestamps to solve the geometric deflection angle locally, without relying on controllable FTL information transfer, and at the operational level does not conflict with the no-communication theorem.
Q8: Will continuous rate variation and drastic flow changes cause DPSP to crash?	Under normal mission geometry and sampling density, design goal is not to crash. However, protocol is discrete-sampling system strictly bounded by Shannon-Nyquist theorem: when $f_{field} > f_{sample}/2$, aliasing must occur, quadratic compensation may misjudge high-frequency noise as low-frequency trend. For this extreme case, system relies on IMU emergency

No. & Question	Answer & Boundary Defense
	trigger to automatically switch to high-frequency sampling (Adaptive Trigger) and a priori model assistance. If beyond design envelope, protocol may still fail. This is physical boundary of discrete system, not design flaw.

Chapter 7: Conclusion

This paper attempts to reframe the issue of special relativity exceeding its real-number domain at extreme speeds as "apparent reverse-order reception of wavefront signals," and proposes the formal framework of "topological causal trunk" and "geometric branch curvature" in TCLM, alongside the Dual-Pulse Synchronization Protocol (DPSP) and its underlying rate-solving algorithm as a conceptual operational framework for engineering implementation.

The model adopts "spacetime worldline continuous-field integration" as its primary working hypothesis for dynamic alignment. The measured value of GPS (+38 μs/day) and the NIST calculated value of the Earth-Mars system (477 μs/day) serve as input calibration cases for relativistic time rates, matching in order of magnitude the time-rate variations in dynamic gravity wells and non-inertial frames required by the protocol; however, this is input calibration, not empirical validation of autonomous reconstruction by the protocol. The protocol's autonomous reconstruction of causal order relative to the initial handshake in dynamic environments remains a theoretical deduction pending end-to-end simulation and empirical validation.

Regarding precision, this paper explicitly distinguishes two levels:

1. **Hardware Resolution Limit**: The local resolution of current optical lattice clocks has reached the 10^{-18} to 10^{-19} order of magnitude, which is a theoretical upper bound of hardware capability, not a solo contribution of the protocol.
2. **Engineering Alignment Error Budget**: In the Earth-Mars case, after multi-frequency dispersion compensation and dynamic metric-field fitting, the alignment error-budget target relative to the initial handshake reference is at the nanosecond level ($\mathcal{O}(10^{-9}\ \mathrm{s})$). This is a budget target; only if subsequent end-to-end simulation and empirical validation pass may it satisfy the stringent requirements of interplanetary deep-space navigation and communication. Cross-galactic scales are a logical extension whose error budget requires separate modeling.

Furthermore, whether this protocol may extend to a distributed "chronometric gravitational-wave response network" remains an early concept without sensitivity and noise-budget assessment. This paper does not claim that it can replace or supplement existing interferometers, nor does it claim that it possesses gravitational-wave detection capability.

This paper primarily addresses two things:

1. **Re-characterization of a mathematical boundary**: The Lorentz transformation yields imaginary (complex) values when $v > c$, which should be understood as "the mathematical model exceeding its real domain," not "the physical flow of time backward."
2. **Proposal of a measurement method**: The Dual-Pulse Synchronization Protocol (DPSP), which uses ex-post comparison of optical-event timestamps (affected by spatial path and local curvature) and a set of local reference-event timestamps (currently envisioned as quantum-entanglement measurement events, whose mechanism differs from optical latency) to separate spatial latency from local time-flow

rates. This protocol does not rely on interpretations of "whether reference events are theoretically synchronous," utilizing solely operable delay-difference measurements.

The remaining chapters serve as engineering calibration, extended applications, and boundary defenses, and do not constitute claims regarding the existence of FTL, gravitational-wave detection capability, FTL quantum-entanglement communication, or the engineering feasibility of cross-galactic scales.

On a conceptual level, TCLM's causal tree attempts to provide a coexisting path for Newton's and Einstein's views on time.

Final Proposition: FTL Is Not a Means to Return to the Past

Under the TCLM framework, the physical limit of speed exceeding c is geometric latency compressed to zero:

$$\lim_{\eta \to 0^+} L_{\text{propagation}} = 0 \quad (L \in [0, \infty))$$

This limit is zero latency ($L = 0$, i.e., Ping=0), not negative latency ($L < 0$). Since $L < 0$ is undefined in the domain, $v > c \Rightarrow t < 0$ is invalid in operational definition.

If a time machine exists, its mechanism must not lie on the path of speed superposition. This path has been closed topologically and measure-theoretically.

Appendix A: Conditional Phenomenological Remarks: Apparent Effects Within the TCLM Framework If Some Class of FTL Mechanism Exists (Not an Engineering Necessity, and Not a Claim for the Existence of FTL)

1. **Conditional Inference 1: Apparent Time-Axis Decoupling and "Apparent Reverse-Order" Observation**
    - **Hypothetical Situation**: If some class of FTL entity or FTL wavefront exists and is described within the TCLM framework as moving toward an observer, then under that model the observer may "first see the arrival image, then receive photons from the origin and midpoint."
    - **Dual-Track Path Mechanism (Model-Internal Classification)**:
        - **Type A (Relative-Distance FTL)**: The rate of increase of the proper distance between two points exceeds c due to changes in the spatial geometry itself. A typical example is that under cosmic expansion, sufficiently distant galaxies recede at speeds exceeding the speed of light. This is a rate of change of relative distance, not an intrinsic speed of an object exceeding c in its local reference frame; it does not involve local FTL motion and does not violate causality. This mechanism can also be equivalently described as "space flow carrying an object": the variation of the metric simultaneously determines the change of distance and the direction of space flow — two languages for the same geometric fact. If the distribution of $T_{\mu\nu}$ can be actively constructed, then in theory the local distance field can be changed; here "theoretically realizable" means that the general-relativistic field equation permits a corresponding mathematical solution, without involving whether the $T_{\mu\nu}$ required by that solution can be constructed from known forms of matter or energy.

        - **Type B (Discrete FTL / Topological Transition, Model-Internal Assumption)**: This type currently has no observational case and is purely a model-internal inference. If realized via a topological jump isomorphic to the "quantum tunneling effect," then under that model the observer may experience a "Discontinuous Frame": the old image at point A has not yet faded, and a new image at point B bursts into existence instantaneously, possibly accompanied by a single high-energy spacetime-metric perturbation radiation.

    - **Causal Defense (Model-Internal, only discussing apparent effects under velocity-related mechanisms)**: Such phenomena are classified in TCLM as apparent wavefront reverse-order reception on geometric branches. On the topological trunk, event settlement is set to be strictly monotonically increasing. Under this speculative model, the observer may receive some future image signals in advance, but this does not constitute substantive interference with the event ontology, nor does it permit sending information to the past by means of speed breakthrough; the grandfather paradox under this path is not permitted within this model. This paper does not claim that any such FTL mechanism exists or is feasible.

2. **Conditional Remark 2: Read-Only Causal Horizon at the Extreme-Speed Limit (Model-Internal Concept)**

    - Under such assumptions, when a carrier approaches and exceeds the speed of light, its geometric branch curvature enters an extreme high-pressure region. Any operation attempting to return a "dynamic control command (write)" to the upstream trunk is, within the model, set as non-writable.
    - Under this speculative model, such a carrier can only play a passive-observer role and cannot become a tool for causal reversal. This paragraph does not constitute a claim regarding FTL engineering feasibility.

Associated Notes

1. Lee Wang Hin (2026). *The Circular Reasoning in the Superluminal Ban: A Logical Note on the Einstein Synchronization Convention*. Zenodo. https://doi.org/10.5281/zenodo.22782047

Declarations

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This study is a purely theoretical physics and logical framework exploration; no experimental datasets were generated or analyzed. All logical premises and thought experiments upon which the derivations are based are detailed in the manuscript text.

AI Participation Statement:
During the writing process, mathematical derivation and formula generation were completed by AI, with

multiple AI systems cross-checking the consistency of logic and formulas; LaTeX typesetting and language polishing were also completed by AI. The core logical framework, theoretical postulates of the Tree-like Causal Latency Model (TCLM), design of the Dual-Pulse Synchronization Protocol (DPSP), and final derived conclusions were independently conceived and completed by the author. The author assumes full responsibility for the entire content and academic integrity of this manuscript.